

1

PROBLEM-SOLVING WITH ALGORITHMS



The act of catching a ball is remarkable. A ball may start so far away that it seems only a speck on the horizon. It may be in the air for only a few short seconds or less. The ball will meet air resistance, wind, and of course, gravity, moving in something like a parabolic arc. And each time a ball is thrown, it is sent with a different force, at a different angle, and in a different environment with different conditions. So how is it that the moment a batter hits a baseball, an outfielder 300 feet away seems to immediately know where to run in order to catch it before it hits the ground?

This question is called *the outfielder problem*, and it's still being discussed in scholarly journals today. We're starting with the outfielder problem because it has two very different solutions: an analytic solution and